Dr. Benjamin Rickles
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Professional Summary

Detail-oriented neuroscience PhD with extensive experience in neurophysiological data analysis, machine learning, and computational modeling. Proven ability to design complex experiments, handle large datasets, and develop analytical pipelines. Passionate about leveraging data-driven insights to advance understanding of cognitive processes. Finds joy in explaining complex concepts to large audiences.

Core Skills & Technologies

- Collaboration and filesharing (Github)
- Python (numpy, scipy, pandas, gensim, scikit-learn)
- MATLAB & R (statistics, data processing)
- Natural Language Processing (NLP) & Computational Modeling
- Machine Learning, Statistical Methods, Data Visualization & Validation
- Experimental Design & Human Subjects Research
- IRB Protocols & Human Subjects Research Compliance

Education

PhD in Neuroscience & Cognitive Science
University of Maryland, 2017–2023

- Relevant coursework: Computational Psycholinguistics, Brain Dynamics & EEG
- Certificate: Introduction to I-Corps

Master of Psychology
University of Oregon, 2016–2017

Bachelor of Arts in Psychology
University of Pittsburgh, 2005–2009

Research & Data Analysis Experience

Graduate Researcher | University of Maryland | 2017–2024

- Designed and implemented neurophysiological experiments, including EEG and ERP studies, involving data acquisition, preprocessing, and analysis for cognitive and language processing.
- Developed pipelines for data analysis using Python (numpy, scipy, gensim, nlptk) and MATLAB, automating preprocessing and statistical evaluation of neurophysiological signals.
- Applied machine learning techniques to EEG/ERP datasets to identify neural correlates of narrative comprehension and semantic processing.
- Conducted large-scale data collection ensuring data quality and integrity.

- Utilized natural language processing to correlate textual stimuli with neural data.

Research Intern | Adobe Inc. | 2020

- Developed algorithms for neural data processing and integration of eye-tracking with EEG in VR environments.
- Analyzed neurophysiological data using Python, applying statistical models and visualization tools.

Additional Research Experience in ERP/fMRI data analysis, experimental design, and neurocognitive modeling across multiple institutions.

Technical & Data-Driven Projects

- Created computational models of language semantics using vector space models (GloVe) to interpret neural signals.
- Built and maintained datasets, pipelines, and repositories (Git, Github) for neurophysiological experiments, ensuring reproducibility.
- Conducted advanced statistical analysis and machine learning classification on neurophysiological data, improving signal detection and interpretability.

Professional Experience *(Selected roles)*

Associate Clinical Account Specialist | Johnson & Johnson MedTech | 2024–Present

- Managed technical support during complex electrophysiology procedures, maintaining focus in high-pressure environments, and fostering collaboration among clinicians.

Evaluation Coordinator | Advanced Education R&D Fund (AERDF) | 2023–2024

- Led scientific validation studies involving data collection and analysis for literacy interventions.

Chief Technology Officer & Innovator | Repurpose Farm Plastic LLC | 2021–2023

- Developed innovative recycling methods, managed stakeholder interviews, and authored research grants demonstrating research and project management expertise.

Teaching & Leadership

- Instructor of record for university-level courses on neuroscience, research methods, and developmental psychology.
- Led workshops in R-Studio, training students in data analysis tools relevant to neuroscience research.

Awards & Recognitions

- Best Poster, 8th Annual John Hopkins Omega Psi Conference
- William Hodos Dissertation Assistantship, NACS, University of Maryland
- Georgia State University Language & Literacy Initiative Graduate Fellowship
- Semi-finalist 2023 DoGood Challenge award recipient for Repurpose Farm Plastic, LLC
- University of Maryland, 2022 Sustainability Research and Innovation Grant Recipient